

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 09 · Momentum, Impulse and Collisions

With Answers

Questions with full worked solutions

- 35 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 09

Momentum, Impulse & Collisions

Your classified route

16

QUESTIONS

101

MARKS

2

FAMILIES

7

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 09 | FAMILY 01

Conservation of momentum in a direct collision

WHAT THIS FAMILY ASKS

Apply signed one-dimensional momentum conservation to two particles.

VARIANT ROUTE

V01 Particles separate after collision

V02 Particles coalesce

V03 Unknown mass/speed parameter or ratio

V04 Unknown post-collision direction

8 classified questions

57 total marks

January 2020 Q1 — complete original question

1. Two particles, P and Q , of mass m_1 and m_2 respectively, are moving on a smooth horizontal plane. The particles are moving towards each other in opposite directions along the same straight line when they collide directly. Immediately before the collision, both particles are moving with speed u .

The direction of motion of each particle is reversed by the collision.

Immediately after the collision, the speed of Q is $\frac{1}{3}u$.

- (a) Find, in terms of m_2 and u , the magnitude of the impulse exerted by P on Q in the collision.

(3)

- (b) Find, in terms of m_1 , m_2 and u , the speed of P immediately after the collision.

(3)

- (c) Hence show that $m_2 > \frac{3}{4}m_1$

(2)

January 2020 Q1 — detailed solution

Particles separate after collision

ALL
8 marks**Choose a positive direction**

Take P's original direction as positive. Before collision, $v_P = u$ and $v_Q = -u$. After collision, Q has velocity $\frac{u}{3}$ because its direction is reversed.

Find the impulse on Q

Impulse on Q equals change in momentum: $I = m_2 \left(\frac{u}{3} - (-u) \right) = m_2 \left(\frac{4u}{3} \right) = \frac{4}{3}m_2u$.

Use conservation of momentum

Let the speed of P after collision be v . Since P reverses direction, its velocity is $-v$. Hence $m_1u - m_2u = -m_1v + \frac{1}{3}m_2u$.

Solve for P's speed

Rearrange: $m_1v = \frac{4}{3}m_2u - m_1u$, so $v = \frac{(4m_2 - 3m_1)u}{3m_1}$.

Use the fact that speed is positive

Since $u > 0$ and $m_1 > 0$, the numerator must be positive: $4m_2 - 3m_1 > 0$. Therefore $m_2 > \frac{3}{4}m_1$.

Final answer (a) $I = \frac{4}{3}m_2u$. (b) speed of P = $\frac{(4m_2 - 3m_1)u}{3m_1}$. (c) $m_2 > \frac{3}{4}m_1$.

January 2023 Q2 — complete original question

2. Two particles, A and B , are moving in a straight line in opposite directions towards each other on a smooth horizontal surface when they collide directly.

Particle A has mass $3m$ kg and particle B has mass m kg.

Immediately before the collision, both particles have a speed of 1.5 m s^{-1}

Immediately after the collision, the direction of motion of A is unchanged and the difference between the speed of A and speed of B is 1 m s^{-1}

(a) Find (i) the speed of A immediately after the collision,

(ii) the speed of B immediately after the collision.

(5)

(b) Find, in terms of m , the magnitude of the impulse exerted on B in the collision.

(3)

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January 2023 Q2 — detailed solution

Particles separate after collision

ALL
8 marks**Choose a positive direction**

Take the original direction of A as positive. Before the collision, A has velocity 1.5 and B has velocity -1.5 .

Use the speed difference

After the collision, A keeps the same direction. Let A's speed be v . The valid motion has B moving in A's direction with speed $v + 1$.

Use conservation of momentum

$$3m(1.5) + m(-1.5) = 3mv + m(v + 1).$$

Find the two speeds

$$4.5m - 1.5m = 3mv + mv + m, \text{ so } 3 = 4v + 1. \text{ Hence } v = 0.5, \text{ and B's speed is } v + 1 = 1.5.$$

Find the impulse on B

$$\text{For B, impulse equals change in momentum: } I = m(1.5 - (-1.5)) = 3m.$$

State the magnitude

The magnitude of the impulse is $3m$ N s.

Final answer (a)(i) speed of A = 0.5 m s^{-1} . (a)(ii) speed of B = 1.5 m s^{-1} . (b) $I = 3m$ N s.

Summer 2023 Q1 — complete original question

1. A particle A has mass 4 kg and a particle B has mass 2 kg .

The particles move towards each other in opposite directions along the same straight line on a smooth horizontal table and collide directly.

Immediately before the collision, the speed of A is $2u\text{ m s}^{-1}$ and the speed of B is $3u\text{ m s}^{-1}$

Immediately after the collision, the speed of B is $2u\text{ m s}^{-1}$

The direction of motion of B is reversed by the collision.

- (a) Find, in terms of u , the speed of A immediately after the collision. (3)
- (b) State the direction of motion of A immediately after the collision. (1)
- (c) Find, in terms of u , the magnitude of the impulse received by B in the collision.
State the units of your answer. (3)

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Summer 2023 Q1 — detailed solution

Particles separate after collision

ALL
7 marks**Choose a positive direction**

Take the original direction of A as positive. Before the collision, A has velocity $2u$ and B has velocity $-3u$. After the collision, B has velocity $2u$ because its direction is reversed.

Use conservation of momentum

Let the velocity of A after the collision be v . Then $4(2u) + 2(-3u) = 4v + 2(2u)$.

Solve for the velocity of A

$8u - 6u = 4v + 4u$, so $2u = 4v + 4u$. Hence $4v = -2u$ and $v = -\frac{u}{2}$.

State the speed and direction

The speed of A is the magnitude of its velocity, so it is $\frac{u}{2}$. The negative sign shows that A has reversed direction.

Find the impulse on B

Impulse equals change in momentum. For B, $I = 2(2u - (-3u)) = 2(5u) = 10u$.

Attach the units

The impulse is $10u$ N s, equivalently $10u$ kg m s⁻¹.

Final answer (a) speed of A = $\frac{u}{2}$. (b) A reverses direction. (c) $I = 10u$ N s.

October 2024 Q1 — complete original question

1. Particle A has mass $4m$ and particle B has mass $3m$.

The particles are moving in opposite directions along the same straight line on a smooth horizontal surface when they collide directly.

Immediately **before** the collision, the speed of A is $2x$ and the speed of B is x .

Immediately **after** the collision, the speed of A is y and the speed of B is $5y$.

The direction of motion of each particle is reversed as a result of the collision.

- (a) Show that $y = \frac{5}{11}x$. (3)
- (b) Find, in terms of m and x , the magnitude of the impulse received by A in the collision. (3)

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October 2024 Q1 — detailed solution

Particles separate after collision

ALL
6 marks**Choose a sign convention**

Take the initial direction of A as positive. Before collision, A has velocity $2x$ and B has velocity $-x$. After collision, both reverse direction, so A has velocity $-y$ and B has velocity $5y$.

Apply conservation of momentum

$4m(2x) + 3m(-x) = 4m(-y) + 3m(5y)$. The left side is $5mx$ and the right side is $11my$.

Show the ratio

$5mx = 11my$, so $\frac{y}{x} = \frac{5}{11}$.

Use change in momentum for A

The impulse received by A is the change in momentum of A: $I = 4m|-y - 2x| = 4m(y + 2x)$.

Substitute y

Using $y = \frac{5x}{11}$, $I = 4m\left(\frac{5x}{11} + 2x\right) = 4m\left(\frac{27x}{11}\right) = \frac{108}{11}mx$.

Final answer (a) $\frac{y}{x} = \frac{5}{11}$. (b) $I = \frac{108}{11}mx$.

Summer 2026 Q1 — complete original question

1. Railway truck A has mass $5m$ and railway truck B has mass $3m$. The trucks are moving towards each other in opposite directions along the same straight track when they collide directly. Immediately before the collision the speed of A is u and the speed of B is $1.5u$

As a result of the collision, the trucks couple together.

- (a) Find, in terms of u , the common speed of the trucks immediately after the collision. (3)
- (b) Find, in terms of m and u , the magnitude of the impulse exerted on B by A in the collision. (3)

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Summer 2026 Q1 — detailed solution

Particles coalesce

ALL
6 marks**(a) Common speed after coupling — Step 1**

Take truck A's initial direction as positive. Then $u_A = u$ and $u_B = -1.5u$.

(a) Common speed after coupling — Step 2

Conservation of momentum gives $5mu + 3m(-1.5u) = 8mv$. Hence $0.5mu = 8mv$, so $v = u/16$ and the sign is positive.

(b) Impulse on truck B — Step 1

For truck B, impulse equals its mass multiplied by its change in velocity. $J_B = 3m\{u/16 - (-3u/2)\} = 3m(25u/16) = 75mu/16$.

(b) Impulse on truck B — Step 2

The positive sign shows that the impulse on B acts in A's initial direction. Therefore the impulse magnitude is $75mu/16$.

Final answer (a) $u/16$ (b) $75mu/16$

January 2022 Q2 — complete original question

2. A particle P has mass km and a particle Q has mass m . The particles are moving towards each other in opposite directions along the same straight line when they collide directly.

Immediately before the collision, P has speed $3u$ and Q has speed u .

As a result of the collision, the direction of motion of each particle is reversed and the speed of each particle is halved.

- (a) Find the value of k .

(4)

- (b) Find, in terms of m and u , the magnitude of the impulse exerted on Q in the collision.

(3)

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January 2022 Q2 — detailed solution

Unknown mass, speed parameter or ratio

ALL
7 marks**Choose a positive direction**

Take the original direction of P as positive. Before the collision, P has velocity $3u$ and Q has velocity $-u$.

Write the velocities after collision

After collision, each direction is reversed and each speed is halved. So P has velocity $-\frac{3}{2}u$ and Q has velocity $\frac{1}{2}u$.

Use conservation of momentum

$$km(3u) + m(-u) = km\left(-\frac{3}{2}u\right) + m\left(\frac{1}{2}u\right).$$

Find k

Divide by mu : $3k - 1 = -\frac{3}{2}k + \frac{1}{2}$. Hence $\frac{9}{2}k = \frac{3}{2}$, so $k = \frac{1}{3}$.

Find impulse on Q

Impulse on Q equals change in momentum of Q: $I = m\left(\frac{1}{2}u - (-u)\right) = \frac{3}{2}mu$.

State the magnitude

The magnitude of the impulse is $\frac{3}{2}mu$.

Final answer (a) $k = \frac{1}{3}$. (b) $I = \frac{3}{2}mu$.

Summer 2021 Q1 — complete original question

1. A particle P has mass $3m$ and a particle Q has mass $5m$. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal surface. The particles collide directly.

Immediately before the collision the speed of P is ku , where k is a constant, and the speed of Q is $2u$.

Immediately after the collision the speed of P is u and the speed of Q is $3u$.

The direction of motion of Q is reversed by the collision.

- (a) Find, in terms of m and u , the magnitude of the impulse exerted on Q by P in the collision.

(2)

- (b) Find the two possible values of k .

(5)

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Summer 2021 Q1 — detailed solution

Unknown post-collision direction

ALL
7 marks**Choose a positive direction**

Take P's original direction as positive. Q initially has velocity $-2u$. Since Q reverses direction and then has speed $3u$, its final velocity is $3u$.

Find the impulse on Q

Impulse equals change in momentum: $I = 5m(3u - (-2u)) = 25mu$.

Use conservation of momentum

Before collision, total momentum is $3m(ku) + 5m(-2u) = 3mku - 10mu$.

Case 1: P continues in its original direction

If P has final velocity u , final momentum is $3m(u) + 5m(3u) = 18mu$. So $3mku - 10mu = 18mu$, giving $3k = 28$ and $k = \frac{28}{3}$.

Case 2: P reverses direction

If P has final velocity $-u$, final momentum is $3m(-u) + 5m(3u) = 12mu$. So $3mku - 10mu = 12mu$, giving $3k = 22$ and $k = \frac{22}{3}$.

Final answer (a) $I = 25mu$. (b) $k = \frac{28}{3}$ or $\frac{22}{3}$.

October 2025 Q2 — complete original question

2.

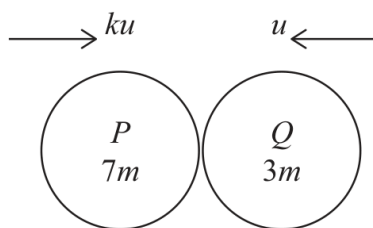


Figure 2

Particle P of mass $7m$ and particle Q of mass $3m$ are moving in opposite directions along the same straight line on a smooth horizontal surface. The particles collide directly.

Immediately **before** the collision, the speed of P is ku and the speed of Q is u , as shown in Figure 2.

Immediately **after** the collision, the speed of P is w and the speed of Q is $2w$.

The direction of motion of Q is reversed by the collision.

The impulse received by Q in the collision has magnitude $\frac{7}{2}mu$.

(a) Find w in terms of u .

(3)

(b) Find the two possible values of k .

(5)

October 2025 Q2 — detailed solution

Unknown post-collision direction

ALL
8 marks**Part (a): Choose a positive direction**

Take right as positive. For Q, the before-velocity is $-u$ and the after-velocity is $+2w$. The impulse on Q is therefore $+\frac{7}{2}mu$.

Part (a): Apply impulse as change in momentum

$\frac{7}{2}mu = 3m(2w - (-u)) = 3m(2w + u)$. Cancel m : $\frac{7}{2}u = 6w + 3u$, so $6w = \frac{1}{2}u$ and $w = \frac{u}{12}$.

Part (b): Write the momentum before collision

Before collision, total momentum is $7m(ku) + 3m(-u) = mu(7k - 3)$.

Part (b): Case 1 — P continues to the right

The final momentum is $7mw + 3m(2w) = 13mw$. Hence $(7k - 3)u = 13w = \frac{13u}{12}$. Therefore $7k = \frac{49}{12}$ and $k = \frac{7}{12}$.

Part (b): Case 2 — P moves to the left

The final momentum is $7m(-w) + 3m(2w) = -mw$. Hence $(7k - 3)u = -w = -\frac{u}{12}$. Therefore $7k = \frac{35}{12}$ and $k = \frac{5}{12}$.

Final answer (a) $w = \frac{u}{12}$. (b) $k = \frac{7}{12}$ or $\frac{5}{12}$.

TOPIC 09 | FAMILY 02

Impulse as change in momentum

WHAT THIS FAMILY ASKS

Use $I = m(v - u)$ with a declared positive direction and distinguish vector/sign from magnitude.

VARIANT ROUTE**V01** Impulse on one particle**V02** Given impulse to recover velocity**V03** Slack string becomes taut impulsively**8 classified questions**

44 total marks

October 2022 Q1 — complete original question

1. A railway truck S of mass 20 tonnes is moving along a straight horizontal track when it collides with another railway truck T of mass 30 tonnes which is at rest. Immediately before the collision the speed of S is 4 m s^{-1}

As a result of the collision, the two railway trucks join together.

Find

- (a) the common speed of the railway trucks immediately after the collision, (2)
- (b) the magnitude of the impulse exerted on S in the collision, stating the units of your answer. (3)

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October 2022 Q1 — detailed solution

Impulse on one particle

ALL
5 marks**Use conservation of momentum**

The trucks join together, so they move with a common speed v . Using tonnes consistently, $20(4) + 30(0) = (20 + 30)v$.

Find the common speed

$80 = 50v$, so $v = 1.6 \text{ m s}^{-1}$.

Use change in momentum of S

Truck S has mass 20 000 kg. Its speed changes from 4 to 1.6 m s^{-1} .

Find the impulse magnitude

$I = 20000|1.6 - 4| = 20000(2.4) = 48000 \text{ N s}$.

Final answer (a) $v = 1.6 \text{ m s}^{-1}$. (b) $I = 48000 \text{ N s}$.

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January 2024 Q2 — complete original question

2.

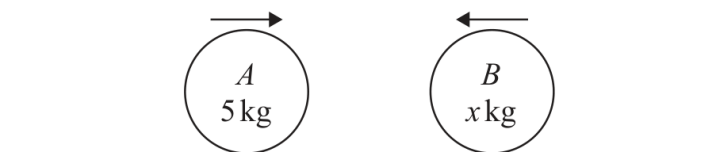
**Figure 2**

Figure 2 shows two particles, A and B , moving in opposite directions on a smooth horizontal surface. Particle A has mass 5 kg and particle B has mass $x\text{ kg}$.

The particles collide directly.

Immediately before the collision, the speed of A is 3 m s^{-1} and the speed of B is $x\text{ m s}^{-1}$

Immediately after the collision, the speed of A is 1 m s^{-1} and its direction of motion is unchanged.

Immediately after the collision, the speed of B is 1.5 m s^{-1}

(a) Find the value of x .

(3)

(b) Find the magnitude of the impulse exerted on A by B in the collision.

(3)

January 2024 Q2 — detailed solution

Impulse on one particle

ALL
6 marks**Choose a sign convention**

Take the initial direction of A as positive. Before collision, A has velocity 3 and B has velocity $-x$.

Set up conservation of momentum

After collision, A still moves in the same direction with velocity 1. B has speed 1.5 in the positive direction for the valid solution. Hence $5(3) + x(-x) = 5(1) + 1.5x$.

Solve for x

$15 - x^2 = 5 + 1.5x$, so $x^2 + 1.5x - 10 = 0$. This gives $x = 2.5$ or $x = -4$. Since mass is positive, $x = 2.5$.

Use change in momentum of A

The impulse exerted on A equals the change in momentum of A: $I = 5(1 - 3) = -10 \text{ N s}$ using the chosen positive direction.

Give the magnitude

The magnitude of the impulse is 10 N s.

Final answer (a) $x = 2.5$. (b) $I = 10 \text{ N s}$.

January 2021 Q2 — complete original question

2. Two particles, P and Q , have masses $2m$ and m respectively. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal plane. The particles collide directly.

Immediately before the collision, the speed of P is $3u$ and the speed of Q is $2u$.

The magnitude of the impulse exerted on Q by P in the collision is $5mu$.

Find

- (a) the speed of P immediately after the collision, (3)
- (b) the speed of Q immediately after the collision. (3)

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January 2021 Q2 — detailed solution

Given impulse to recover velocity

ALL
6 marks**Choose a positive direction**

Take P's original direction as positive. Before collision, P has velocity $3u$ and Q has velocity $-2u$.

Use impulse on P

The impulse on P is opposite to P's original motion and has magnitude $5mu$. Let P's velocity after collision be v . Then $-5mu = 2m(v - 3u)$.

Find P speed

Divide by m : $-5u = 2v - 6u$, so $2v = u$ and $v = \frac{u}{2}$.

Use impulse on Q

The impulse on Q is in P's original direction. Let Q's velocity after collision be w . Then $5mu = m(w - (-2u))$.

Find Q speed

$5u = w + 2u$, so $w = 3u$. Therefore the speed of Q is $3u$.

Final answer (a) speed of P = $\frac{u}{2}$. (b) speed of Q = $3u$.

Summer 2022 Q1 — complete original question

1. Two particles, P and Q , are moving towards each other in opposite directions along the same straight line when they collide directly. Immediately before the collision the speed of Q is $2u$. The mass of Q is $3m$ and the magnitude of the impulse exerted by P on Q in the collision is $4mu$.

Find

- (a) the speed of Q immediately after the collision, (3)

- (b) the direction of motion of Q immediately after the collision. (1)

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Summer 2022 Q1 — detailed solution

Given impulse to recover velocity

ALL
4 marks**Use impulse as change in momentum**

Take the original direction of Q as positive. Q has mass $3m$ and initial velocity $2u$. Let its velocity after collision be v .

Set up the impulse equation

The impulse has magnitude $4mu$ and acts opposite to Q's original motion, so $4mu = 3m(2u - v)$.

Find the speed

Cancel m : $4u = 6u - 3v$. Thus $3v = 2u$ and $v = \frac{2u}{3}$.

State the direction

Since v is positive in Q's original direction, Q continues in the same direction after the collision.

Final answer (a) speed = $\frac{2u}{3}$. (b) same direction as before the collision.

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Summer 2025 Q1 — complete original question

1.

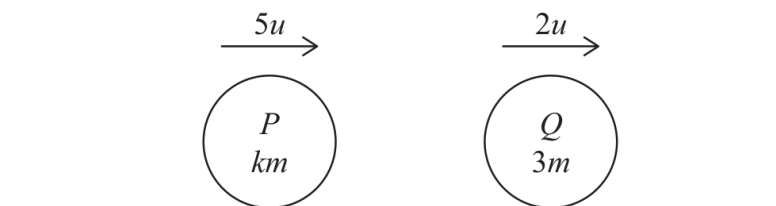


Figure 1

Figure 1 shows two particles, P and Q , moving in the same direction along the same straight line on a smooth horizontal surface.

Particle P has mass km and particle Q has mass $3m$

The particles collide directly.

Immediately before the collision, the speed of P is $5u$ and the speed of Q is $2u$

Immediately after the collision, the speed of P is $2u$ and its direction of motion is unchanged.

Immediately after the collision, the speed of Q is v

The impulse exerted on Q by P in the collision has magnitude $4.5mu$

(a) Find v in terms of u only.

(3)

(b) Find the value of k

(3)

Summer 2025 Q1 — detailed solution

Given impulse to recover velocity

ALL
6 marks**Part (a): Read the impulse direction**

Take the direction of motion as positive. P approaches Q from behind, so the impulse on Q acts forwards. For Q, mass = $3m$, before-velocity = $2u$ and after-velocity = v .

Part (a): Apply impulse as change in momentum

$4.5mu = 3m(v - 2u)$. Cancel m : $4.5u = 3v - 6u$, so $3v = 10.5u$ and $v = 3.5u = \frac{7u}{2}$.

Part (b): Write momentum before and after

Before collision, total momentum is $km(5u) + 3m(2u) = 5kmu + 6mu$. After collision, it is $km(2u) + 3m\left(\frac{7u}{2}\right) = 2kmu + 10.5mu$.

Part (b): Solve for k

$5k + 6 = 2k + 10.5$, so $3k = 4.5$ and $k = 1.5 = \frac{3}{2}$.

Final answer (a) $v = \frac{7u}{2}$. (b) $k = \frac{3}{2}$.

January 2026 Q1 — complete original question

1. A particle P of mass $3m$ and a particle Q of mass $5m$ are on a smooth horizontal surface. The particles move towards each other in opposite directions along the same straight line and collide.

Immediately before the collision, the speed of P is $3u$ and the speed of Q is $4u$

The magnitude of the impulse exerted on P by Q in the collision is $\frac{33}{2}mu$

Find

- (i) the speed of P immediately after the collision,
- (ii) the speed of Q immediately after the collision.

(6)

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January 2026 Q1 — detailed solution

Given impulse to recover velocity

ALL
6 marks**Choose a sign convention**

Take the initial direction of P as positive. Then before impact, $v_P = 3u$ and $v_Q = -4u$. The impulse on P acts opposite to P's initial motion, so its signed impulse is $-\frac{33}{2}mu$.

Use impulse on P

For P, impulse equals change in momentum: $-\frac{33}{2}mu = 3m(v_P - 3u)$. Dividing by $3m$ gives $v_P - 3u = -\frac{11u}{2}$, so $v_P = -\frac{5u}{2}$.

Interpret the speed of P

The negative sign means P has reversed direction. Its speed is therefore $\left| -\frac{5u}{2} \right| = \frac{5u}{2}$.

Use the equal and opposite impulse on Q

Q receives impulse $+\frac{33}{2}mu$ in the positive direction. Hence $\frac{33}{2}mu = 5m(v_Q - (-4u)) = 5m(v_Q + 4u)$.

Interpret the speed of Q

Dividing by $5m$ gives $v_Q + 4u = \frac{33u}{10}$, so $v_Q = -\frac{7u}{10}$. Q is still moving in its original direction, and its speed is $\frac{7u}{10}$.

Final answer (i) speed of P = $\frac{5u}{2}$. (ii) speed of Q = $\frac{7u}{10}$.

October 2019 Q1 — complete original question

1. Two particles, P and Q , have masses $3m$ and $2m$ respectively. The particles are connected by a light inextensible string. Initially P and Q are at rest on a smooth horizontal plane with the string slack.

Particle P is then projected along the plane directly away from Q with speed $4u$. At the same instant, particle Q is projected along the plane in the opposite direction with speed $3u$.

Find

- (a) the common speed of the particles immediately after the string becomes taut, (3)
- (b) the magnitude of the impulse exerted on Q at the instant when the string becomes taut. (3)

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October 2019 Q1 — detailed solution

Slack string becomes taut impulsively

ALL
6 marks**Choose a direction**

Take P's initial direction as positive. Then P initially has velocity $4u$ and Q initially has velocity $-3u$.

Use conservation of momentum

When the string becomes taut, both particles move with common speed v . Hence $3m(4u) + 2m(-3u) = (3m + 2m)v$.

Find the common speed

$$12mu - 6mu = 5mv, \text{ so } v = \frac{6u}{5}.$$

Use impulse on Q

Impulse on Q equals change in momentum: $I = 2m \left(\frac{6u}{5} - (-3u) \right).$

Find the impulse magnitude

$$I = 2m \left(\frac{6u}{5} + \frac{15u}{5} \right) = 2m \cdot \frac{21u}{5} = \frac{42}{5}mu.$$

Final answer (a) $v = \frac{6u}{5}$. (b) $I = \frac{42}{5}mu$.

Summer 2024 Q1 — complete original question

1. Two particles, A and B , have masses m and $3m$ respectively. The particles are connected by a light inextensible string. Initially A and B are at rest on a smooth horizontal plane with the string slack.

Particle A is then projected along the plane away from B with speed U .

Given that the common speed of the particles immediately after the string becomes taut is S

- (a) find S in terms of U . (2)
- (b) Find, in terms of m and U , the magnitude of the impulse exerted on A immediately after the string becomes taut. (3)

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Summer 2024 Q1 — detailed solution

Slack string becomes taut impulsively

ALL
5 marks**Use conservation of momentum**

When the string becomes taut, A and B move with the same speed S . Before this, only A is moving. Hence $mU = (m + 3m)S = 4mS$.

Find the common speed

From $mU = 4mS$, cancel m to get $S = \frac{U}{4}$.

Use impulse on A

The impulse on A equals the magnitude of its change in momentum: $I = m \left| \frac{U}{4} - U \right|$.

Simplify the impulse

$$I = m \left(\frac{3U}{4} \right) = \frac{3mU}{4}.$$

Final answer (a) $S = \frac{U}{4}$. (b) $I = \frac{3mU}{4}$.